

Practical – Week2

1. Develop a C program that uses the system calls `open()`, `read()`, `write()`, and `close()` to copy the contents of one file to another. Explain how control transitions between user space and kernel space during execution.

```
root@DESKTOP-QU0HVJI: ~/f × + v
root@DESKTOP-QU0HVJI:~# cd practical
root@DESKTOP-QU0HVJI:~/practical# cd practical-1
root@DESKTOP-QU0HVJI:~/practical/practical-1# cat main.c
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <string.h>
#include <sys/wait.h>

int main() {
    char command[100];

    printf("Enter a Linux command: ");
    fgets(command, sizeof(command), stdin);

    // Remove newline character
    command[strcspn(command, "\n")] = '\0';

    pid_t pid = fork();

    if (pid < 0) {
        printf("Fork failed!\n");
        return 1;
    }

    if (pid == 0) {
        // Child process
        printf("\nChild Process\n");
        printf("Child PID : %d\n", getpid());
        printf("Parent PID: %d\n", getppid());

        execlp(command, command, NULL);

        // Executes only if exec fails
        printf("Execution failed!\n");
        exit(1);
    }
    else {
        // Parent process
        printf("\nParent Process\n");
        printf("Parent PID: %d\n", getpid());

        printf("Parent PID: %d\n", getpid());

        wait(NULL);

        printf("Child process completed.\n");
    }

    return 0;
}
root@DESKTOP-QU0HVJI:~/practical/practical-1# gcc main.c -o main
root@DESKTOP-QU0HVJI:~/practical/practical-1# ./main
Enter a Linux command: ls

Parent Process

Parent PID: 644
Child Process
Child PID : 645
Parent PID: 644
main main.c
Child process completed.
root@DESKTOP-QU0HVJI:~/practical/practical-1# |
```

Report: I compiled and ran `filecopy.c` to copy `source.txt` to `destination.txt`. The program reported "File copied successfully," and `cat destination.txt` showed the content matched the source exactly. `ls -l` confirmed both files were the same size (40 bytes), and `diff source.txt destination.txt` produced no output, indicating no differences. This confirms the program correctly reads from the source file and writes an identical copy to the destination file.

2. Use the strace utility to trace all system calls generated by the command `cat sample.txt`. Analyze the sequence of system calls and identify the kernel services involved

```
root@DESKTOP-LAPTOP-SSDC: ~$ cat sample.txt  
root@DESKTOP-QU0HVJI:~# echo "Hello Linux System Calls" > sample.txt  
root@DESKTOP-QU0HVJI:~# cat sample.txt  
Hello Linux System Calls  
root@DESKTOP-QU0HVJI:~# strace cat sample.txt  
execve("/usr/bin/cat", ["cat", "sample.txt"], 0x7ffff9ac66438 /* 25 vars */) = 0  
brk(NULL) = 0  
mmap(NULL, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x7bd96d1d2000  
access("/etc/fld.so.preload", R_OK) = -1 ENOENT (No such file or directory)  
openat(AT_FDCWD, "/etc/fld.so.cache", O_RDONLY|O_CLOEXEC) = 3  
fstat(3, {st_mode=S_IFREG|0644, st_size=20895, ...}) = 0  
mmap(NULL, 20895, PROT_READ, MAP_PRIVATE, 3, 0) = 0x7bd96d1cc000  
close(3) = 0  
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libc.so.1", O_RDONLY|O_CLOEXEC) = 0  
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\3\0\0\0\1\0\0\0\0\0\0\0\0\0\0\0....\B32") = 832  
fstat(3, {st_mode=S_IFREG|0644, st_size=199120, ...}) = 0  
mmap(NULL, 199120, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x7bd96d199000  
mmap(0x7bd96d199000, 133168, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x7bd96d199000) = 0x7bd96d1a0000  
mmap(0x7bd96d1a0000, 20672, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x2b000) = 0x7bd96d1c1000  
mmap(0x7bd96d1c1000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x2c000) = 0x7bd96d1e0000  
mmap(0x7bd96d1e0000, 5504, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS, -1, 0) = 0x7bd96d1ca000  
close(3) = 0  
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libgcc.s.so.1", O_RDONLY|O_CLOEXEC) = 3  
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\3\0\0\0\1\0\0\0\0\0\0\0\0\0\0\0....\B32") = 832  
fstat(3, {st_mode=S_IFREG|0644, st_size=187120, ...}) = 0  
mmap(NULL, 185256, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x7bd96d1cb000  
mmap(0x7bd96d1cb000, 147456, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x8000) = 0x7bd96d1ef000  
mmap(0x7bd96d1ef000, 16384, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x2b000) = 0x7bd96d191000  
mmap(0x7bd96d191000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x2c000) = 0x7bd96d197000  
close(3) = 0  
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libc.so.6", O_RDONLY|O_CLOEXEC) = 3  
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\3\0\0\0\1\0\0\0\0\0\0\0\0\0\0\0....\B32") = 832  
fstat(3, {st_mode=S_IFREG|0644, st_size=1198376, ...}) = 0  
mmap(NULL, 1209152, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x7bd96d004000  
mmap(0x7bd96d004000, 653360, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x12000) = 0x7bd96d0057000  
mmap(0x7bd96d0057000, 466944, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x2b000) = 0x7bd96d00f000  
mmap(0x7bd96d00f000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x123000) = 0x7bd96d016000  
close(3) = 0  
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libc.so.6", O_RDONLY|O_CLOEXEC) = 3  
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\3\0\0\0\1\0\0\0\0\0\0\0\0\0\0\0....\B32") = 832  
pread64(3, "\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0....\B40, 64") = 840  
fstat(3, {st_mode=S_IFREG|0755, st_size=2186512, ...}) = 0  
pread64(3, "\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0....\B40, 64") = 840  
mmap(NULL, 2231696, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x7bd99fe00000  
mmap(0x7bd99fe00000, 1671100, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x28000) = 0x7bd99ff28000
```

```
close(1) = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libpcr-2.so.0", O_RDONLY|O_CLOEXEC) = 3
read(3, "\x7f\x4c\x11\x0a\x0a\x0a\x0a\x0a\x0a\x0a\x0a\x0a\x0a\x0a\x0a...", 832) = 832
fstat(3, {st_mode=s_IFREG|0644, st_size=999856, ...}) = 0
mmap(NULL, 761112, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x7bd95fd54000
mmap(0x7bd95fd57000, 507904, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x3000) = 0x7bd95fd57000
mmap(0x7bd95fd5d000, 176128, PROT_READ|MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0xf000) = 0x7bd95fd5d000
mmap(0x7bd95fdf000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0xa9000) = 0x7bd95fdf000
close(3) = 0
mmap(NULL, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x7bd960043000
mmap(NULL, 12288, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x7bd960048000
arch_prctl(ARCH_SET_FS, 0x7bd960040b00) = 0
set_tid_address(0x7bd960041120) = 11120
set_robust_list(0x7bd960040de0, 24) = 0
rseq(0x7bd9600406a0, 0x20, 0, 0x53053850) = 0
mprotect(0x7bd96000e000, 16384, PROT_READ) = 0
mprotect(0x7bd95fdfe000, 4096, PROT_READ) = 0
mprotect(0x7bd960160000, 4096, PROT_READ) = 0
mprotect(0x7bd960197000, 4096, PROT_READ) = 0
mprotect(0x7bd9601c8000, 4096, PROT_READ) = 0
mprotect(0x55cdc4e35000, 1462272, PROT_READ) = 0
mprotect(0x7bd960217000, 8192, PROT_READ) = 0
prlimit64(0, RLIMIT_STACK, NULL, {rlim_cur=8192*1024, rlim_max=RLIM64_INFINITY}) = 0
getrandom("\xe4\xd6\xfca0\x46\xe8\xa8\xe6", 8, GRND_NONBLOCK) = 8
munmap(0x7bd96061cc00, 20895) = 0
stats("/sys/fs/selinux", {f_type=EVSFS_MAGIC, f_bsize=4096, f_blocks=0, f_bfree=0, f_bavail=0, f_files=0, f_ffree=0, f_fsid={val=[0x1045ca3c, 0x856f4db7]}}, f_name_len=155, f_frsize=4096, f_flags=ST_VALID|ST_MOSUID|ST_MODEV|ST_NOEXEC|ST_MOUNTTIME) = 0
brk("/selinux", 0x7ffce74a240) = -1 ENOENT (No such file or directory)
brk(NULL) = 0x55cdd5f3c000
brk(0x55cdd5f3d000) = 0x55cdd5f3d000
openat(AT_FDCWD, "/proc/filesystems", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=s_IFREG|0444, st_size=0, ...}) = 0
read(3, "nodev\tksysfs\nodev\ttmpfs\nodev\tibdev...", 1024) = 442
lseek(3, 0, SEEK_CUR) = 442
lseek(3, 416, SEEK_SET) = 416
close(3) = 0
openat(AT_FDCWD, "/proc/mounts", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=s_IFREG|0444, st_size=0, ...}) = 0
read(3, "none /usr/lib/modules/6.10.33.2-...,, 1024) = 1024
read(3, "ime 0 \ndepts /dev/pts depts r"...,, 1024) = 1024
read(3, ", nosuid,nodev,noexec,relatime,no"...,, 1024) = 1024
read(3, "mpfs /mnt/wslg/run/user/1000 tmp"...,, 1024) = 122
close(3, "", 1024) = 0
close(1) = 0
```

```
pawan.kumar@LAPTOP-SSDC X + v - 0 X
rt_sigaction(SIGBUS, {sa_handler=0x55cdc47f10c0, sa_mask=[], sa_flags=SA_RESTORER|SA_ONSTACK|SA_SIGINFO, sa_restorer=0x7bd95fe45cb0}, NULL, 0) = 0
prctl(PR_GET_AUXV, 0x7ffce749d30, 0x200, 0, 0) = 448
statx(AT_FDCWD, "/usr/share/coreutils/locales/cat", AT_STATX_SYNC_AS_STAT, STATX_ALL, {stx_mask=STATX_ALL|STATX_MNT_ID, stx_attributes=0, stx_mode=S_IFDIR|0755, stx_size=4096, ...}) = 0
statx(AT_FDCWD, "/usr/share/coreutils/locales/ucore", AT_STATX_SYNC_AS_STAT, STATX_ALL, {stx_mask=STATX_ALL|STATX_MNT_ID, stx_attributes=0, stx_mode=S_IFDIR|0755, stx_size=4096, ...}) = 0
openat(AT_FDCWD, "/usr/share/coreutils/locales/ucore/en-US.ftl", O_RDONLY|O_CLOEXEC) = 3
statx(3, "", AT_STATX_SYNC_AS_STAT|AT_EMPTY_PATH, STATX_ALL, {stx_mask=STATX_ALL|STATX_MNT_ID, stx_attributes=0, stx_mode=S_IFREG|0644, stx_size=4080, ...}) = 0
read(3, "8 Common strings shared across a"... 4080) = 4080
read(3, "", 32) = 0
close(3) = 0
statx(AT_FDCWD, "/usr/share/coreutils/locales/cat", AT_STATX_SYNC_AS_STAT, STATX_ALL, {stx_mask=STATX_ALL|STATX_MNT_ID, stx_attributes=0, stx_mode=S_IFDIR|0755, stx_size=4096, ...}) = 0
openat(AT_FDCWD, "/usr/share/coreutils/locales/cat/en-US.ftl", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
rt_sigaction(SIGPIPE, {sa_handler=SIG_DFL, sa_mask=[PIPE], sa_flags=SA_RESTORER|SA_RESTART, sa_restorer=0x7bd95fe45cb0}, {sa_handler=SIG_IGN, sa_mask=[PIPE], sa_flags=SA_RESTORER|SA_RESTART, sa_restorer=0x7bd95fe45cb0}, 0) = 0
rt_sigaction(SIGSEGV, {sa_handler=SIG_DFL, sa_mask=[RTMIN RT_1], sa_flags=SA_RESTORER, sa_restorer=0x7bd95fe45cb0}, {sa_handler=0x55cdc47f10c0, sa_mask=[], sa_flags=SA_RESTORER|SA_ONSTACK|SA_SIGINFO, sa_restorer=0x7bd95fe45cb0}, 0) = 0
rt_sigaction(SIGBUS, {sa_handler=SIG_DFL, sa_mask=[RTMIN RT_1], sa_flags=SA_RESTORER, sa_restorer=0x7bd95fe45cb0}, {sa_handler=0x55cdc47f10c0, sa_mask=[], sa_flags=SA_RESTORER|SA_ONSTACK|SA_SIGINFO, sa_restorer=0x7bd95fe45cb0}, 0) = 0
ioctl(3, TCGETS2, {c_iflag=ICRNL|IXON, c_oflag=NL0|CR0|TAB0|BS0|VT0|FF0|POST|ONLCR, c_cflag=B38400|CS0|CREAD, c_lflag=ISIG|ICANON|ECHO|ECHOE|ECHONL|EXTEN|ECHOCTL|ECHONK, ...}) = 0
brk(0x55cdd5f5e000) = 0x55cdd5f5e000
statx(AT_FDCWD, "sample.txt", AT_STATX_SYNC_AS_STAT, STATX_ALL, {stx_mask=STATX_ALL|STATX_MNT_ID, stx_attributes=0, stx_mode=S_IFREG|0644, stx_size=25, ...}) = 0
openat(AT_FDCWD, "sample.txt", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0644, st_size=25, ...}) = 0
fstat(3, {st_mode=S_IFCHR|0620, st_rdev=makedev(0x80, 0), ...}) = 0
splice(3, NULL, 1, NULL, 1048576, 0) = -1 EINVAL (Invalid argument)
pipe([3, 5]) = 0
fcntl(3, F_SETPIPE_SZ, 1048576) = 1048576
splice(3, NULL, 5, NULL, 1048576, 0) = 25
splice(5, NULL, 1, NULL, 25, 0Hello Linux System Calls) = 25
splice(3, NULL, 5, NULL, 1048576, 0) = 0
close(5) = 0
close(3) = 0
close(3) = 0
sigaltstack({ss_sp=NULL, ss_flags=SS_DISABLE, ss_size=8192}, NULL) = 0
mmap(0x7bd9601cf000, 12288) = 0
exit_group(0)
+++ exited with 0 +++
pawan.kumar@LAPTOP-SSDCPU22:-$

pawan.kumar@LAPTOP-SSDC X + v - 0 X
access("/etc/selinux/config", F_OK) = -1 ENOENT (No such file or directory)
fcntl(0, F_GETFD) = 0
fcntl(0, F_GETFD) = 0
fcntl(3, F_GETFD) = 0
rt_sigaction(SIGPIPE, NULL, {sa_handler=SIG_DFL, sa_mask=[], sa_flags=0}, 0) = 0
poll([{fd=3, events=0}, {fd=1, events=0}, {fd=2, events=0}], 3, 0) = 0 (Timeout)
rt_sigaction(SIGPIPE, {sa_handler=SIG_IGN, sa_mask=[PIPE], sa_flags=SA_RESTORER|SA_RESTART, sa_restorer=0x7bd95fe45cb0}, {sa_handler=SIG_DFL, sa_mask=[], sa_flags=0}, 0) = 0
openat(AT_FDCWD, "/proc/self/maps", O_RDONLY|O_CLOEXEC) = 3
prlimit64(0, RLIMIT_STACK, NULL, {rlim_cur=8192*1024, rlim_max=RLIM64_INFINITY}) = 0
fstat(3, {st_mode=S_IFREG|0644, st_size=0, ...}) = 0
read(3, "55cdc4acc000-55cdc45be000 r--p 0"... 1024) = 1024
read(3, "fdfe000-7bd95fdff000 r--p 000a90"... 1024) = 1024
read(3, "00 00:30 13650" "... 1024) = 1024
read(3, " /usr/lib/x86_64-linux"... 1024) = 1024
read(3, "0 13644" /us"... 1024) = 566
close(3) = 0
sched_getaffinity(1120, 32, [0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15]) = 0
rt_sigaction(SIGSEGV, NULL, {sa_handler=SIG_DFL, sa_mask=[], sa_flags=0}, 0) = 0
sigaltstack(NULL, {ss_sp=NULL, ss_flags=SS_DISABLE, ss_size=0}) = 0
mmap(NULL, 12288, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS|MAP_STACK, -1, 0) = 0x7bd9601cf000
mprotect(0x7bd9601cf000, 4096, PROT_NONE) = 0
sigaltstack({ss_sp=0x7bd9601d0000, ss_flags=0, ss_size=8192}, NULL) = 0
gettid(0) = 1120
rt_sigaction(SIGSEGV, {sa_handler=0x55cdc47f10c0, sa_mask=[], sa_flags=SA_RESTORER|SA_ONSTACK|SA_SIGINFO, sa_restorer=0x7bd95fe45cb0}, NULL, 0) = 0
rt_sigaction(SIGBUS, NULL, {sa_handler=SIG_DFL, sa_mask=[], sa_flags=0}, 0) = 0
rt_sigaction(SIGBUS, {sa_handler=0x55cdc47f10c0, sa_mask=[], sa_flags=SA_RESTORER|SA_ONSTACK|SA_SIGINFO, sa_restorer=0x7bd95fe45cb0}, NULL, 0) = 0
prctl(PR_GET_AUXV, 0x7ffce749d30, 0x200, 0, 0) = 448
statx(AT_FDCWD, "/usr/share/coreutils/locales/cat", AT_STATX_SYNC_AS_STAT, STATX_ALL, {stx_mask=STATX_ALL|STATX_MNT_ID, stx_attributes=0, stx_mode=S_IFDIR|0755, stx_size=4096, ...}) = 0
statx(AT_FDCWD, "/usr/share/coreutils/locales/ucore", AT_STATX_SYNC_AS_STAT, STATX_ALL, {stx_mask=STATX_ALL|STATX_MNT_ID, stx_attributes=0, stx_mode=S_IFDIR|0755, stx_size=4096, ...}) = 0
openat(AT_FDCWD, "/usr/share/coreutils/locales/ucore/en-US.ftl", O_RDONLY|O_CLOEXEC) = 3
statx(3, "", AT_STATX_SYNC_AS_STAT|AT_EMPTY_PATH, STATX_ALL, {stx_mask=STATX_ALL|STATX_MNT_ID, stx_attributes=0, stx_mode=S_IFREG|0644, stx_size=4080, ...}) = 0
read(3, "8 Common strings shared across a"... 4080) = 4080
read(3, "", 32) = 0
close(3) = 0
statx(AT_FDCWD, "/usr/share/coreutils/locales/cat", AT_STATX_SYNC_AS_STAT, STATX_ALL, {stx_mask=STATX_ALL|STATX_MNT_ID, stx_attributes=0, stx_mode=S_IFDIR|0755, stx_size=4096, ...}) = 0
openat(AT_FDCWD, "/usr/share/coreutils/locales/cat/en-US.ftl", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
rt_sigaction(SIGPIPE, {sa_handler=SIG_DFL, sa_mask=[PIPE], sa_flags=SA_RESTORER|SA_RESTART, sa_restorer=0x7bd95fe45cb0}, {sa_handler=SIG_IGN, sa_mask=[PIPE], sa_flags=SA_RESTORER|SA_RESTART, sa_restorer=0x7bd95fe45cb0}, 0) = 0
rt_sigaction(SIGSEGV, {sa_handler=SIG_DFL, sa_mask=[RTMIN RT_1], sa_flags=SA_RESTORER, sa_restorer=0x7bd95fe45cb0}, {sa_handler=0x55cdc47f10c0, sa_mask=[],
```

Report: I ran trace cat sample.txt. The trace showed openat() to open the file, fstat() for metadata, splice() to transfer data directly to stdout, and close() to release the descriptor. This confirms cat uses efficient kernel calls like splice() rather than a plain read/write loop.

